Geologic Report

Mini Practical Example

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Executive Summary

This report presents a practical example of a geological study, including location, lithology, structural geology, mineralization, and geochemical analysis. The report provides an educational model for geology and mining students.

1. Location and Accessibility

The studied area is located approximately 25 km northwest of Example City. It is accessible by paved and dirt roads. Geographic coordinates: 24°30'N, 45°10'E.

2. Regional Geology

The area belongs to the Arabian Shield and consists mainly of Precambrian volcanic and plutonic rocks intruded by granitic bodies. The regional geology is characterized by NE-SW structural trends.

3. Lithology

Rock Type	Description

Granite	Pink to grey, coarse-grained, rich in quartz and feldspar.
Basalt	Dark, fine-grained, with vesicles filled by secondary minerals.
Schist	Foliated, medium-grade metamorphic rock with mica and quartz.

4. Structural Geology

The area is affected by several fault systems trending NE-SW and NW-SE. Folds are also observed in schist units, indicating regional compression.

5. Mineralization

Gold mineralization occurs in quartz veins cutting across schist and basalt. Sulphide minerals such as pyrite and chalcopyrite are associated with the veins.

6. Geochemical Analysis

Sample	Au (ppm)	Cu (ppm)	Fe (%)
S1	2.1	250	5.4
S2	0.8	180	4.7
S3	3.5	320	6.2

7. Conclusion

The studied area contains promising gold-bearing quartz veins hosted in metamorphic and volcanic rocks. Further detailed exploration including geophysical surveys and drilling is recommended.